PROJECT DOCUMENTATION

Full Stack Development with MERN

STREAMLINING IT PROCUREMENT: AUTOMATING STANDARD LAPTOP ORDERS WITH FLOW DESIGNER

1. INTRODUCTION

Project Title

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Team Members

Name	Role
Jasmine R	Project Lead
MONISHA A	ServiceNow Developer
Dheensha Hasin B	Workflow Designer
Rathina Selvi R	Tester

2. PROJECT OVERVIEW

Purpose

The purpose of this project is to automate the standard laptop procurement process using **ServiceNow Flow Designer**.

The system helps employees submit laptop requests and automatically manages validation, approval, procurement task creation, notifications, and order tracking.

Features

The main features of the project are:

- Standard laptop request submission.
- Employee information capture.
- Request validation.
- Laptop availability checking.
- Automated manager approval.
- Automatic procurement task creation.
- Automated notifications.
- Laptop order status tracking.
- Request completion.
- Procurement dashboard and reporting.

3. ARCHITECTURE

ServiceNow Platform

The project is implemented using the **ServiceNow platform**. The main components are the request interface, ServiceNow tables, Flow Designer, approval actions, notification system, and procurement task management.

Request Layer

Employees submit a standard laptop procurement request by providing the required employee and laptop information.

Flow Designer

ServiceNow Flow Designer acts as the automation layer. It controls the workflow based on triggers, actions, conditions, and approvals.

The main flow is:

Request Submitted → Validate Request → Check Availability → Manager Approval → Create Procurement Task → Send Notification → Track Order → Complete Request

Database / Data Storage

The project uses **ServiceNow tables** to store and manage procurement information.

Important information includes:

- Employee details
- Laptop request details
- Approval status
- Procurement task
- Order status
- Notification status

Architecture Diagram

Employee

↓

Laptop Request

↓

ServiceNow Platform

↓

Flow Designer

↓

Validation

↓

Manager Approval

↓

Procurement Task

↓

Order Tracking

↓

Notification

↓

Request Completion

4. SETUP INSTRUCTIONS

Prerequisites

The following are required:

- ServiceNow account/instance.
- Access to ServiceNow Flow Designer.
- Appropriate ServiceNow roles and permissions.
- Modern web browser such as Google Chrome or Microsoft Edge.
- Basic knowledge of ServiceNow.

Installation / Configuration

Since the project is developed inside ServiceNow, there is no traditional software installation.

Steps:

1. Log in to the ServiceNow instance.
2. Open the required application or workspace.
3. Create/configure the laptop request table or form.

4. Create the required fields.
5. Open **Flow Designer**.
6. Create a new flow.
7. Configure the trigger for a new laptop request.
8. Add validation and condition actions.
9. Add the approval action.
10. Add procurement task creation.
11. Configure notification actions.
12. Test and activate the flow.

5. PROJECT STRUCTURE

The project is organized into the following ServiceNow components:

Request Form

Used by employees to submit standard laptop requests.

Request Table

Stores employee and laptop request information.

Flow Designer

Contains the automated procurement workflow.

Approval

Handles manager approval or rejection.

Procurement Task

Stores and manages procurement activities.

Notifications

Sends automatic updates to employees and relevant IT personnel.

Dashboard

Displays procurement request and order information.

6. RUNNING THE APPLICATION

The project is executed through the **ServiceNow instance**.

Steps

1. Log in to the ServiceNow instance.
2. Navigate to the laptop procurement application.
3. Open the laptop request form.
4. Enter the required employee and laptop details.
5. Submit the request.
6. Flow Designer automatically starts the workflow.
7. The request is validated.
8. The request is sent for approval.
9. After approval, a procurement task is created.
10. The employee receives a notification.
11. The procurement team updates the order status.
12. The request is completed after delivery.

7. WORKFLOW DOCUMENTATION

The project does not use a conventional REST API as its primary interface. The main functionality is implemented through **ServiceNow forms, tables, Flow Designer actions, approvals, tasks, and notifications**.

Main Workflow Actions

Step	Action
1	Create laptop request
2	Validate request
3	Check laptop availability
4	Request manager approval
5	Process approval decision
6	Create procurement task
7	Send notification
8	Update order status
9	Complete request

8. AUTHENTICATION AND AUTHORIZATION

The project uses the authentication and access-control mechanisms provided by **ServiceNow**.

Users log in to the ServiceNow instance using their authorized accounts.

Access to different activities can be controlled using user roles and permissions.

For example:

- **Employee:** Submit and view their laptop requests.
- **Manager:** Approve or reject requests.
- **IT Administrator:** Manage requests and workflow.
- **Procurement Team:** Process laptop orders.
- **IT Manager:** Monitor procurement information and reports.

9. USER INTERFACE

The project interface includes the following screens:

1. Laptop Request Form

Allows employees to enter their details and laptop requirements.

2. Request List

Displays submitted laptop procurement requests.

3. Approval Screen

Allows the manager to approve or reject a request.

4. Procurement Task

Displays the task created for the procurement team.

5. Order Tracking

Displays the current status of the laptop order.

6. Dashboard

Provides an overview of laptop procurement activities.

Screenshots to include:

- Screenshot 1 – Laptop Request Form
- Screenshot 2 – Submitted Request
- Screenshot 3 – Flow Designer
- Screenshot 4 – Approval
- Screenshot 5 – Procurement Task
- Screenshot 6 – Notification
- Screenshot 7 – Order Status
- Screenshot 8 – Dashboard

10. TESTING

The workflow is tested using different scenarios to verify that each component functions correctly.

Test Case	Expected Result	Status
Submit valid request	Request is created	Pass
Submit incomplete request	Validation identifies missing information	Pass
Check laptop availability	Availability is checked	Pass
Approve request	Procurement flow continues	Pass
Reject request	Request is marked rejected	Pass
Create procurement task	Task is created automatically	Pass
Send notification	Notification is generated	Pass
Update order	Order status is updated	Pass
Complete order	Request is marked completed	Pass

11. SCREENSHOTS / DEMO

The project demonstration should show the complete workflow:

Employee Login

↓

Laptop Request Form

↓

Submit Request

↓

Flow Designer Trigger

↓

Request Validation

↓

Manager Approval

↓

Procurement Task

↓

Notification

↓

Order Tracking

↓

Completed Request

Project Demo Link

To be added: _____

12. KNOWN ISSUES

The following limitations may be present in the current version:

- The workflow depends on correct ServiceNow configuration.
- Laptop availability must be maintained accurately.
- Notifications require proper configuration.
- Changes to procurement rules may require modifications to the flow.
- Access to certain features depends on ServiceNow roles and permissions.

13. FUTURE ENHANCEMENTS

The project can be enhanced with the following features:

1. **Automated inventory management** to check real-time laptop stock.
2. **Vendor integration** for sending orders directly to suppliers.
3. **Automatic purchase order generation** after approval.
4. **Delivery tracking** for monitoring shipment status.
5. **Mobile notifications** for employees and managers.
6. **Advanced analytics dashboard** for procurement performance.
7. **AI-based laptop recommendations** based on employee requirements.
8. **Asset Management integration** to automatically assign the delivered laptop.
9. **Automated escalation** for delayed approvals.
10. **Procurement reports** for analyzing order volume and processing time.